

Module 3

CHY1009 – Chemistry and Environmental Studies

Electrochemistry and Chemical Sensors

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Energy Storage Devices

Energy density vs Power density

Energy density

- **Energy density** is the amount of energy in a given mass (or volume)
- If a system has a high energy density then it is able to store a lot of energy in a small amount of mass.
- A high energy density does not necessarily mean a high power density. An object with a high energy density, but low power density can perform work for a relatively long period of time.
- ***Example of this type of energy storage is a mobile phone.***
- Its power will last most of the day, but to recharge the device, it must be connected to another power source for an hour or more.

Energy density vs Power density

Power density

- **Power density** is the amount of power in a given mass.
- If a system has a high power density, then it can output large amounts of energy based on its mass.
- For example, a **tiny capacitor** may have the same power output as a large battery.
- However, since the capacitor is so much smaller, it has a higher power density.
- Since they release their energy quickly, high power density systems can also recharge quickly.
- An example application of this type of energy storage is a **camera flash**. It has to be small enough to fit inside the camera (or cell phone) but have a high enough power output to light up the subject of your photo. This makes a system with a high power density ideal.

Examples

The starting battery needs to deliver a high power to crank the engine, but not for very long. Therefore it needs to be built with a **high power density, but not necessarily a very high energy density** as the starter motor is only engaged for a few seconds.

The 'house' battery, on the other hand, has the opposite conditions. It will be expected to deliver its energy over a much longer period of time- perhaps a few days use. But the current draw is much less. This means **energy density must be high, but power density can be low**.



Energy density vs Power density

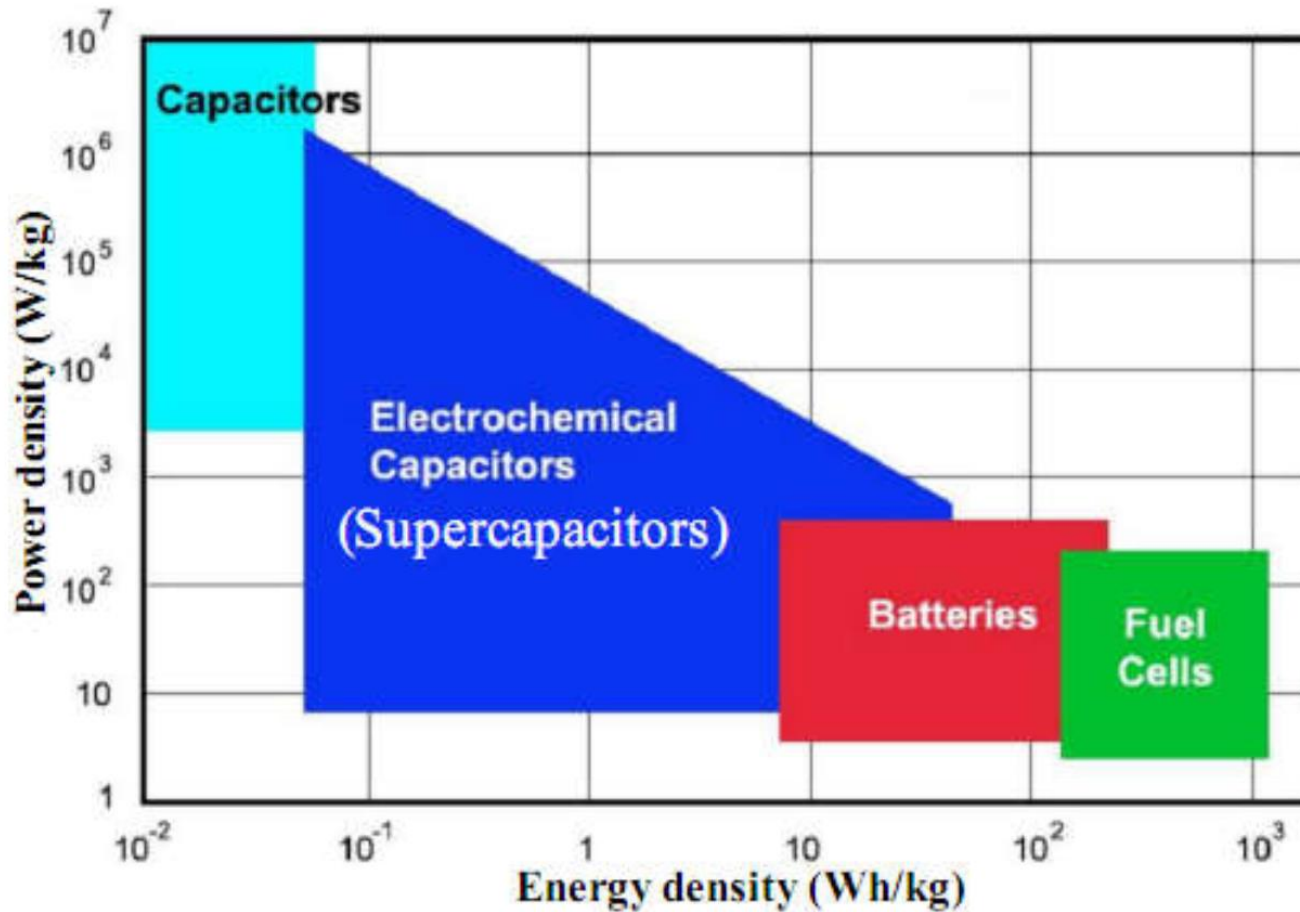


Fig.2. Power density vs. energy density of various energy storage systems.

BATTERY

- A battery is an electrical device consisting of **one or more galvanic cells**, with external connections provided to allow the battery to power electrical devices.

Best commercial batteries:

- ✓ it should be light and compact
- ✓ Its voltage should not vary appreciably during its use
- ✓ It should provide power for a longer period
- ✓ It should be rechargeable

Classification

Primary Cell

Dry cells

Mercury cells

**Primary Cell (or)
Non-rechargeable cells**

**It cannot be returned to their
original state by recharging**

Secondary Cell

Lead storage battery

Lithium ion cell

**Secondary Cell (or)
Rechargeable cells**

It can be returned to their original state by the application of an external electric current, which reverses the oxidation-reduction half cell reactions

Fuel Cell

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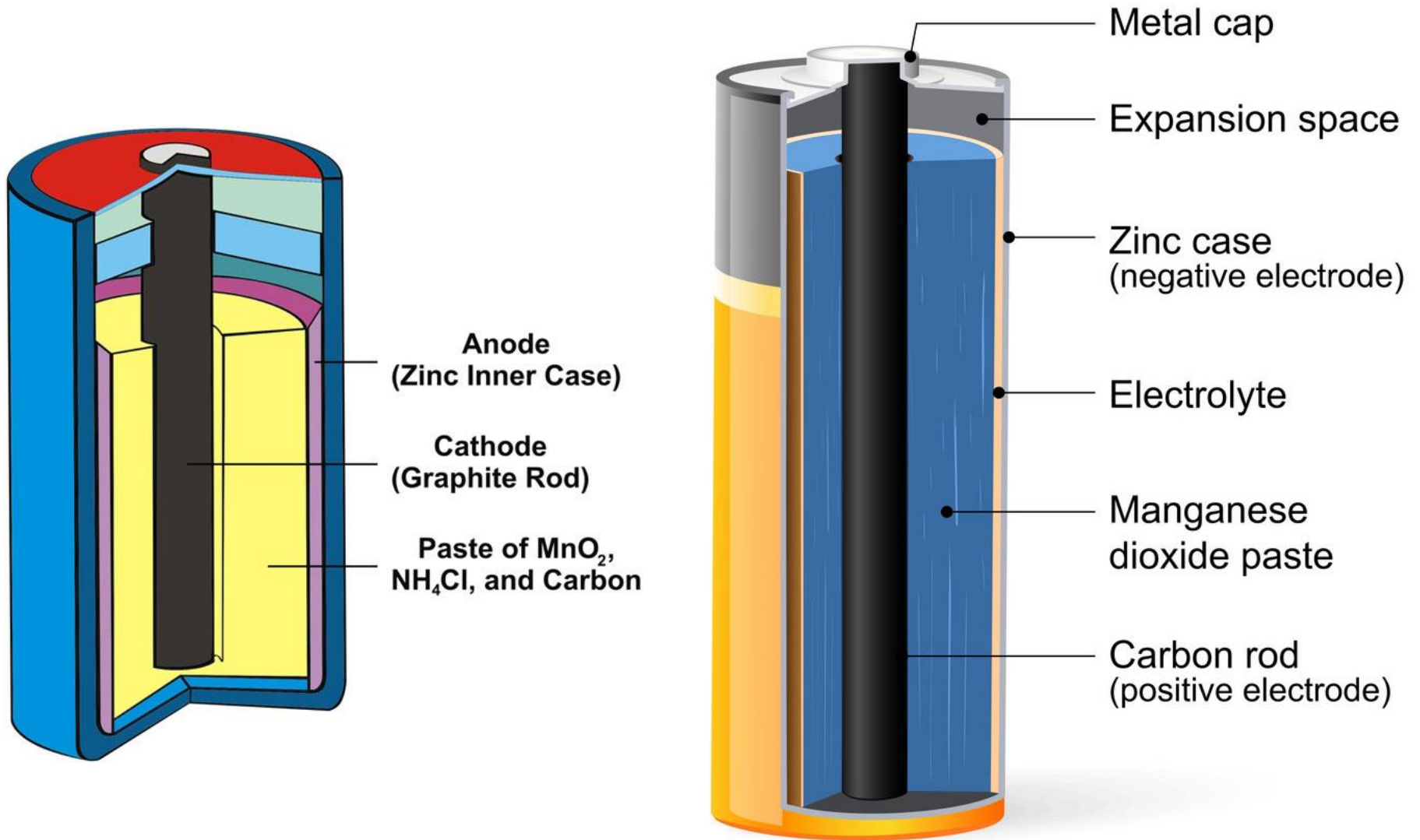
Lead storage battery

Lithium ion cell

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Dry Cell



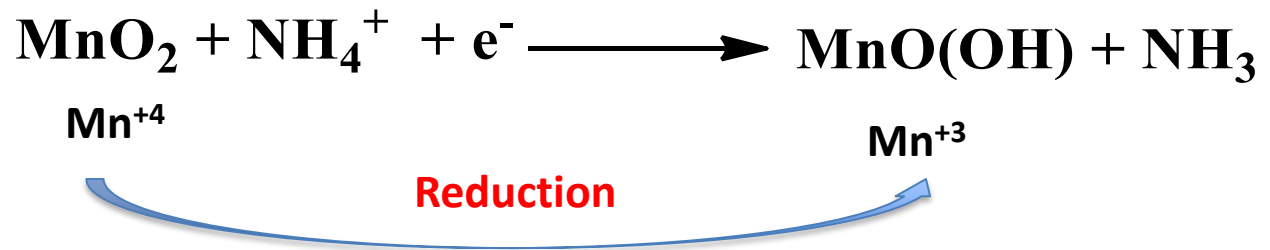
Dry Cell

Anode: Zinc cap

Cathode: Graphite rod

- ✓ Graphite rod is surrounded by the moist paste of MnO_2 , NH_4Cl and carbon

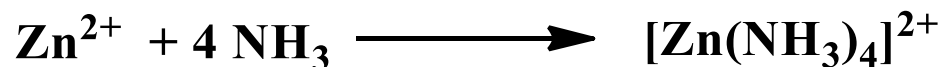
at Cathode



at Anode

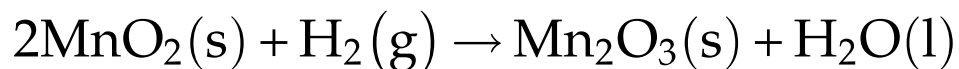
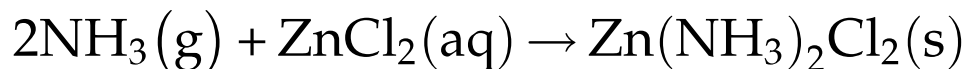


Electrolyte

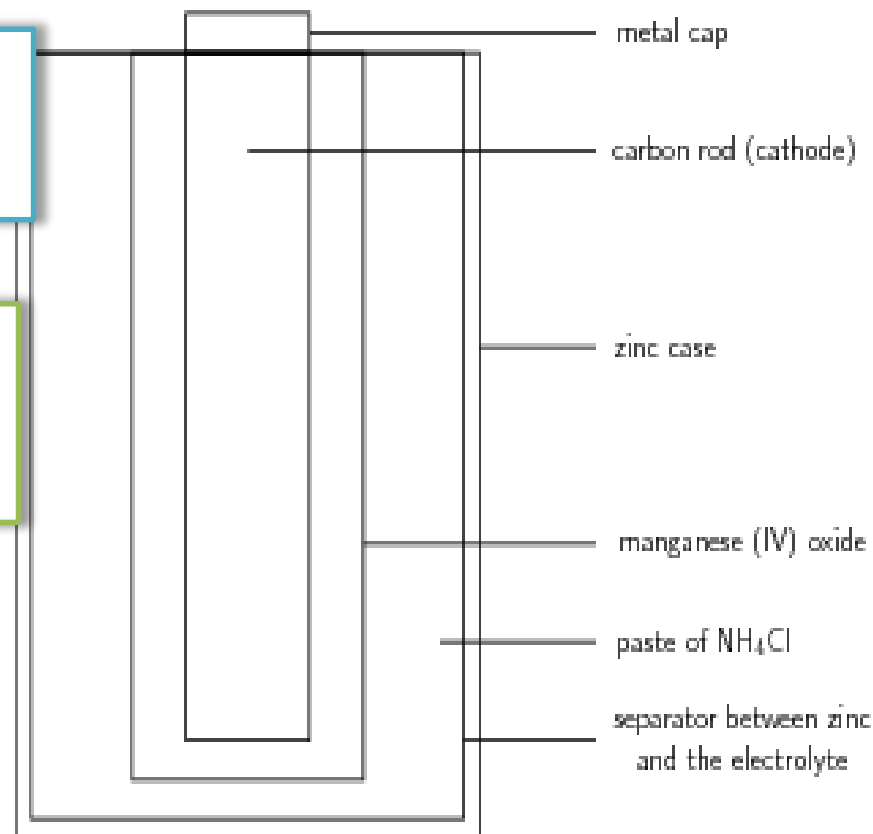


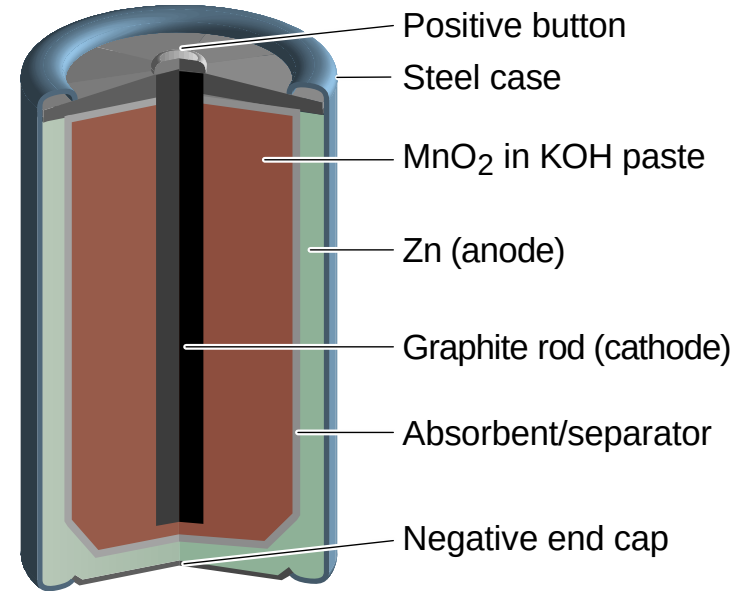
Tetrammine zinc(II) cation

When the electrolyte is a liquid, usually an aqueous salt, the battery is called a wet cell and when it is in the form of a paste or a solid the battery is called a dry cell.



Cell potential (E_{cell}) = 1.5 V

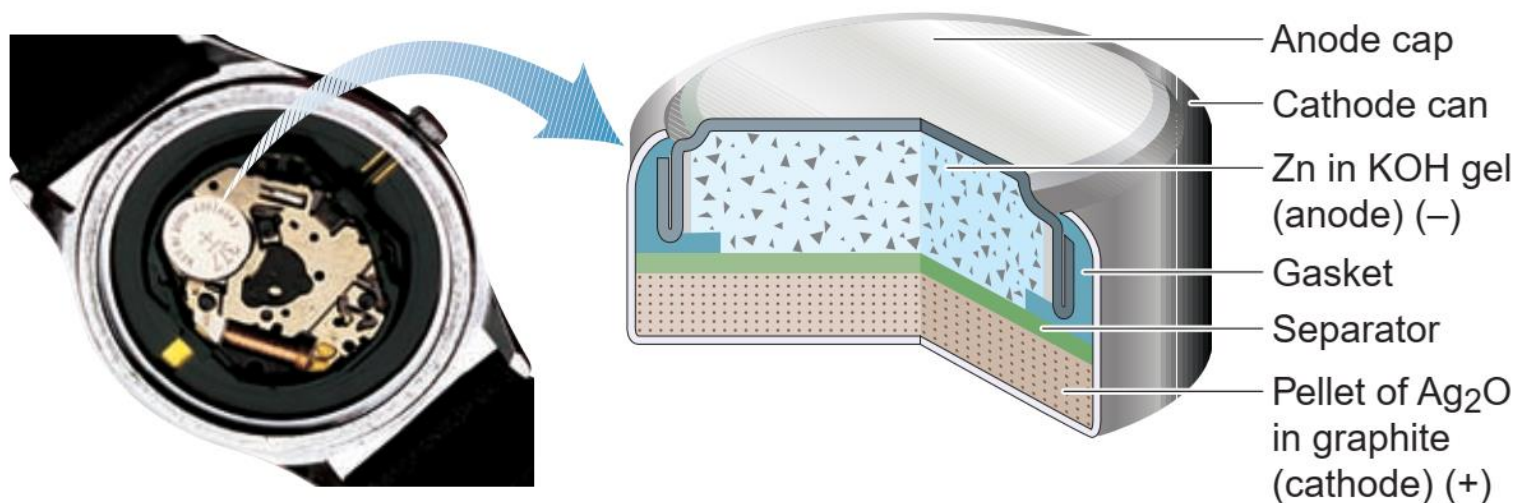




- Alkaline battery powers portable radios, toys, flashlights, and so on, is safe, and comes in many sizes.
- It has no voltage drop, longer shelf life, and better performance in terms of power capability and stored energy

Mercury or Silver (Button) Batteries

- Mercury and silver batteries are quite similar. Both use a **zinc** container as the **anode** (reducing agent) in a basic medium.
- The mercury battery employs HgO as the oxidizing agent, the silver uses Ag_2O , and both use a steel can around the cathode.
- The solid reactants are compacted with KOH and separated with moist paper.



Mercury cells

- Mercury cells are mainly used in watches, video camera, hearing aids, calculators, etc.

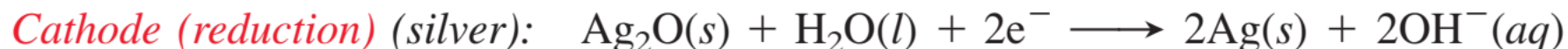
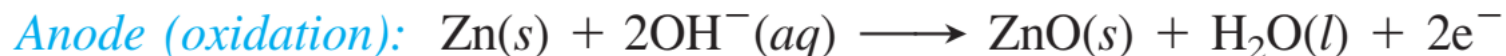
Anode: Zinc or Zinc mercury amalgam

Cathode: Mercuric oxide (HgO)

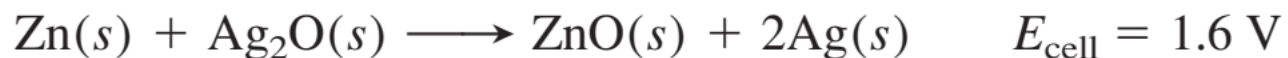
Electrolyte: Potassium hydroxide (KOH)

Since mercuric oxide is a non-conductor, some graphite is mixed with it

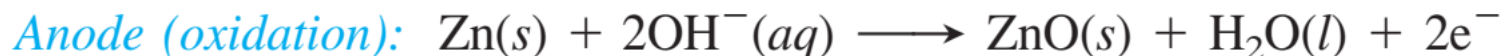
The half-reactions for silver battery:



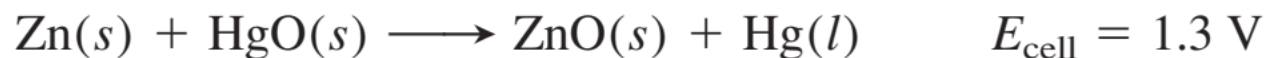
Overall (cell) reaction (silver):



The half-reactions for mercury battery:



Overall (cell) reaction (mercury):



- advantages of a long shelf life of up to 10 years and steady voltage output.

Secondary Batteries (Rechargeable)

- In contrast to primary batteries, a *secondary*, or *rechargeable*, battery is recharged when it runs down *by supplying electrical energy to reverse the cell reaction* and re-form reactant.
- In other words, in this type of battery, the voltaic cells are periodically converted to electrolytic cells to restore *non-equilibrium* concentrations of the cell components.
- By far the most widely used secondary battery is the common **car battery**.
- Two newer types are the **nickel–metal hydride battery** and the **lithium-ion battery**, a secondary version of a lithium battery.

Lead-Acid Battery

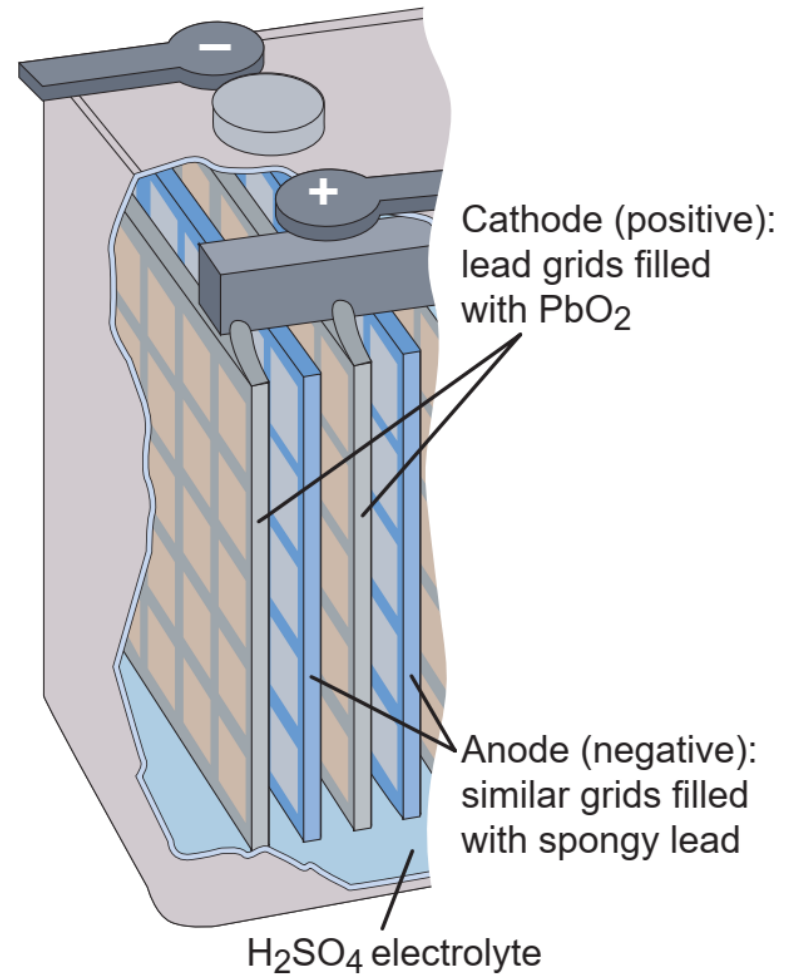


Figure 21.17 Lead-acid battery.

Lead-Acid Battery

- A typical lead-acid car battery has **six cells connected** in series, each of which delivers about 2.1 V for a total of about **12 V**.
- Each cell contains two lead grids loaded with the electrode materials: high-surface-area (spongy) **Pb in the anode** and high-surface-area **PbO₂ in the cathode**.
- The grids are immersed in an **electrolyte** solution of $\sim 4.5\text{ M}$ **H₂SO₄**.

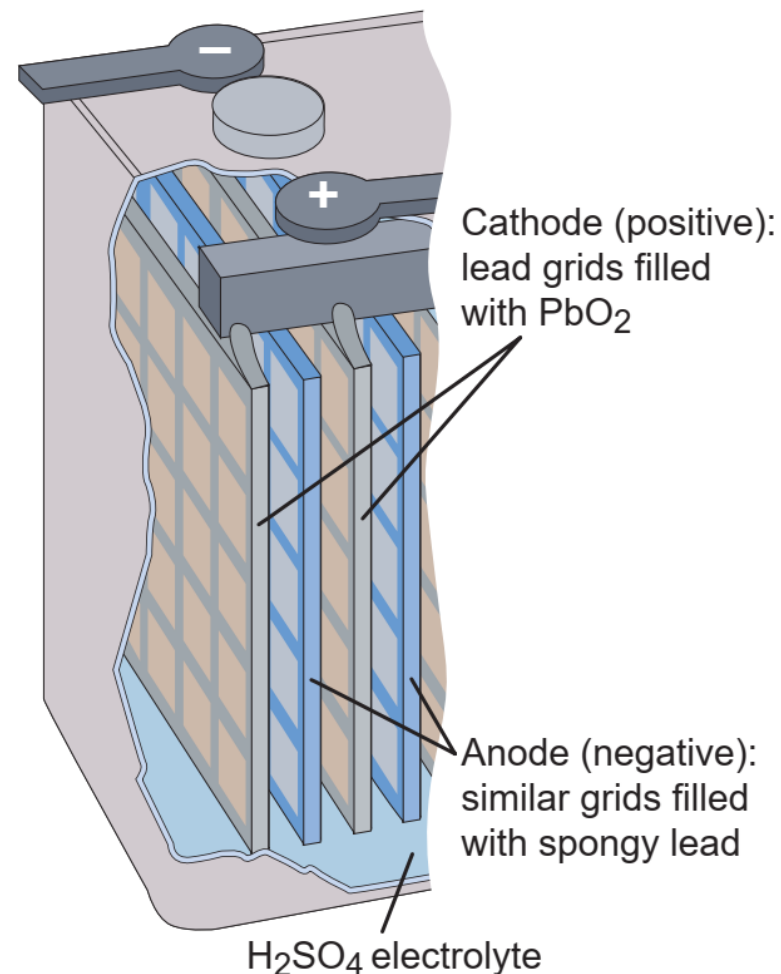
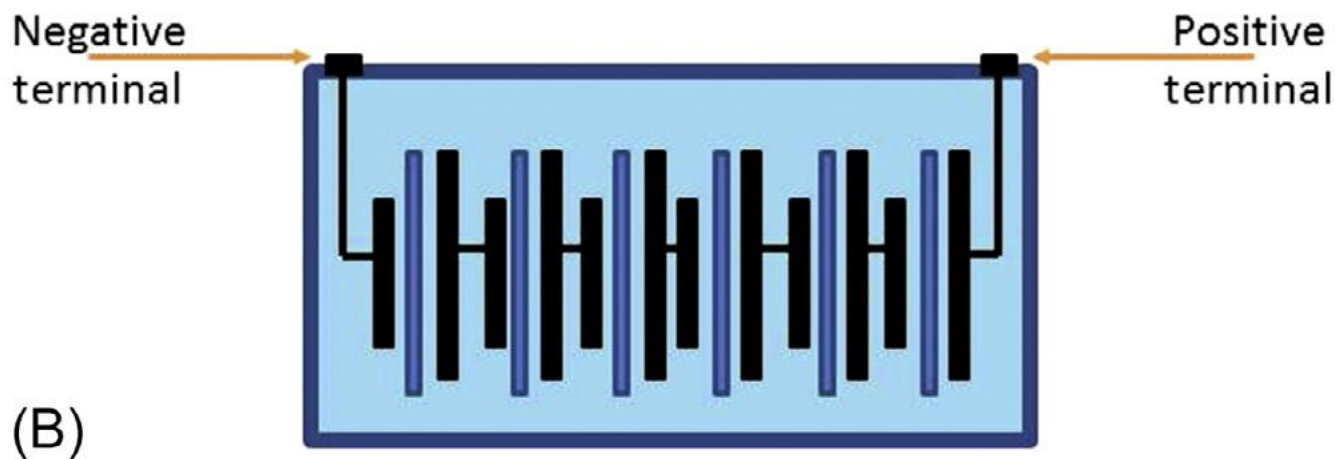
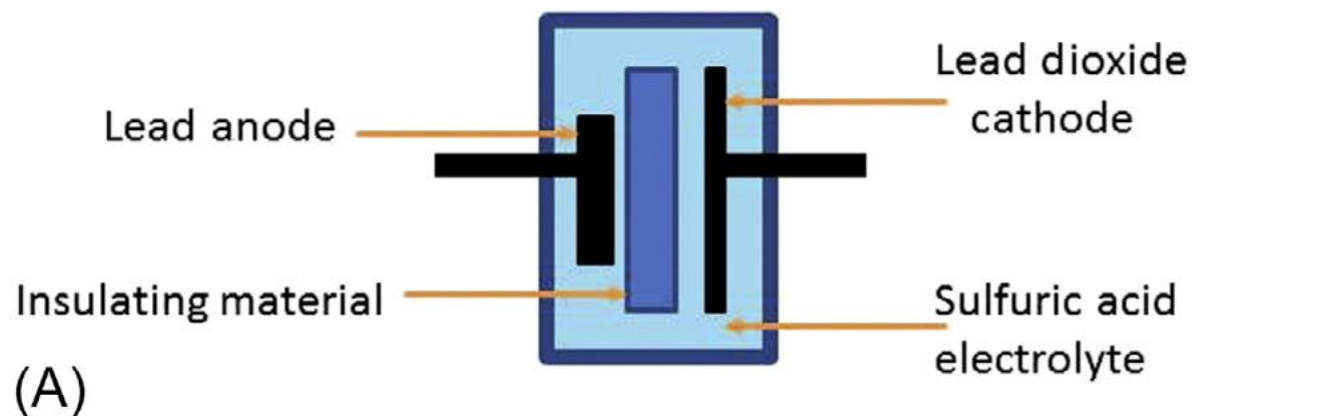
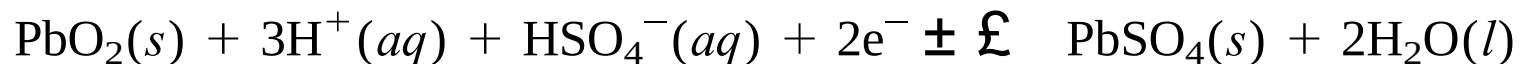


Figure 21.17 Lead-acid battery.

Anode (oxidation):



Cathode (reduction):



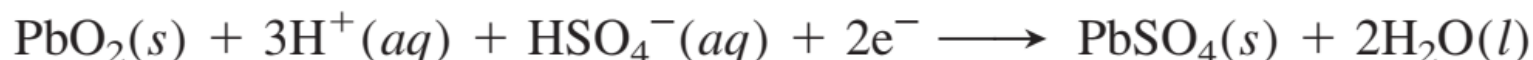
Lead-Acid Battery

- When the cell discharges, it generates electrical energy as a voltaic cell:

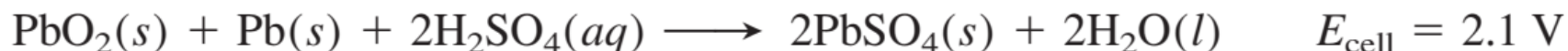
Anode (oxidation):



Cathode (reduction):



Overall (cell) reaction (discharge):

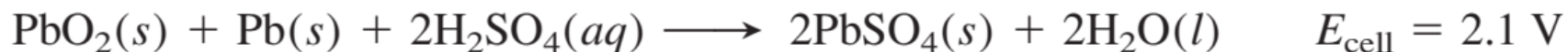


- Notice that both half-reactions produce **Pb²⁺** ions, one through the **oxidation of Pb**, the other through the **reduction of PbO₂**.
- The Pb²⁺ forms PbSO₄(s) at both electrodes by reacting with HSO₄⁻

Lead-Acid Battery

For Discharge:

Overall (cell) reaction (discharge):



When the cell is recharged, it uses electrical energy as an electrolytic cell, and the half-cell and overall reactions are reversed

For Recharge:

Overall (cell) reaction (recharge):



Advantages of lead acid batteries

- **Comparative advantages in performance:**
 - The battery with the longest industrial production time and the most mature technology has **stable, reliable performance and good applicability**;
 - Dilute sulfuric acid is used as electrolyte without flammability. The battery is designed at normal pressure or low pressure with **good safety**;
 - High working voltage, **wide operating temperature range**, suitable for hybrid electric vehicle (HEV) and other **high rate discharge applications**;
 - The large-capacity battery technology is mature and can be made into batteries with **thousands of ampere-hours**, providing convenience for large-scale energy storage.
- **Comparative advantage in cost:** Lead-acid batteries are the cheapest secondary batteries, with unit energy costs about a third of those of lithium-ion or nickel-hydrogen batteries.
- **Comparative advantages of recycling:** Lead-acid battery has simple composition, mature regeneration technology and high recovery value. It is the easiest battery for recycling and reusing.

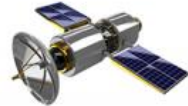
Disadvantages of lead acid batteries

- **Low energy density:** Traditional lead-acid batteries are low in quality and volume energy density. The energy density is only about $\frac{1}{3}$ of that of lithium ion batteries and $\frac{1}{2}$ of that of nickel-hydrogen batteries.
- **Short cycle life:** The cycle life of traditional lead-acid batteries is short, and the theoretical cycle times are about $\frac{1}{3}$ of that of lithium-ion batteries.
- **Lead pollution risks exist in the industrial chain:** Lead is the main raw material of lead-acid batteries. Lead accounts for more than 60% of the battery quality. Lead used in lead-acid batteries accounts for more than 80% of the total lead used globally. Lead is a heavy metal, and the lead-acid battery manufacturing industry chain (including primary lead smelting, battery manufacturing, battery recycling, and secondary lead smelting) has a high risk of lead pollution.
- **Other safety concerns:** Older batteries had a cap on each cell for monitoring electrolyte density and replacing water lost on overcharging. During recharging, some water could be electrolyzed to H_2 and O_2 , which could explode.

Lithium ion battery



Mining



Space Applications



Consumer Electronics



Power Invertors



Energy Storage System



Lithium Ion Batteries



Electric Vehicles
(EV, HEV, PHEV)



Defence /Military Applications



Marines and Submarines



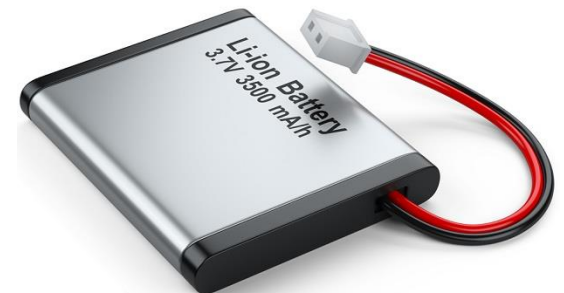
Telecom Towers



Solar and Wind Energy Storage



Railways



The Nobel Prize in Chemistry 2019



Ill. Niklas Elmehed. © Nobel Media.

John B. Goodenough

Prize share: 1/3



Ill. Niklas Elmehed. © Nobel Media.

M. Stanley
Whittingham

Prize share: 1/3



Ill. Niklas Elmehed. © Nobel Media.

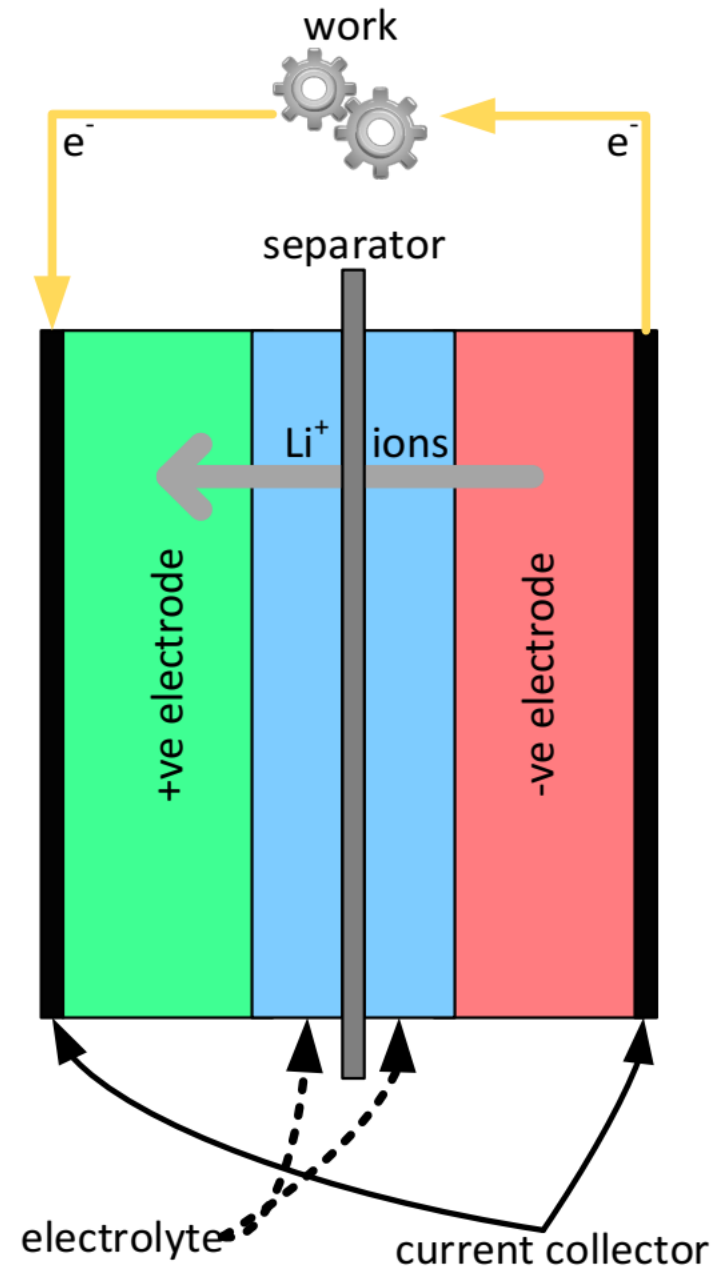
Akira Yoshino

Prize share: 1/3

"for the development of lithium-ion batteries."

Lithium-Ion Battery

- The secondary lithium-ion battery has an **anode** of **Li atoms that lie between sheets of graphite** (designated Li_xC_6).
- The **cathode** is a **lithium metal oxide**, such as LiMn_2O_4 or LiCoO_2
- A typical **electrolyte** is 1 M LiPF_6 (Lithium hexafluorophosphate) in an organic solvent, such as dimethyl carbonate (often mixed with methylethyl carbonate).



Why 'Lithium'

- Position in electrochemical series:
Lowest reduction potential value (or)
highest oxidation potential value

- Position in periodic Table: **Small Size**

Electrochemical series

$\text{F}_2(\text{g}) + 2 \text{e}^-$	$\longrightarrow 2 \text{F}(\text{aq})$	2.87
$\text{H}_2\text{O}_2(\text{aq}) + 2 \text{H}^+(\text{aq}) + 2 \text{e}^-$	$\longrightarrow 2 \text{H}_2\text{O}(\text{l})$	1.78
$\text{MnO}_4^-(\text{aq}) + 8 \text{H}^+(\text{aq}) + 5 \text{e}^-$	$\longrightarrow \text{Mn}^{2+}(\text{aq}) + 4 \text{H}_2\text{O}(\text{l})$	1.51
$\text{Cl}_2(\text{g}) + 2 \text{e}^-$	$\longrightarrow 2 \text{Cl}^-(\text{aq})$	1.36
$\text{Cr}_2\text{O}_7^{2-}(\text{aq}) + 14 \text{H}^+(\text{aq}) + 6 \text{e}^-$	$\longrightarrow 2 \text{Cr}^{3+}(\text{aq}) + 7 \text{H}_2\text{O}(\text{l})$	1.33
$\text{O}_2(\text{g}) + 4 \text{H}^+(\text{aq}) + 4 \text{e}^-$	$\longrightarrow 2 \text{H}_2\text{O}(\text{l})$	1.23
$\text{Br}_2(\text{l}) + 2 \text{e}^-$	$\longrightarrow 2 \text{Br}^-(\text{aq})$	1.09
$\text{Ag}^+(\text{aq}) + \text{e}^-$	$\longrightarrow \text{Ag}(\text{s})$	0.80
$\text{Fe}^{3+}(\text{aq}) + \text{e}^-$	$\longrightarrow \text{Fe}^{2+}(\text{aq})$	0.77
$\text{O}_2(\text{g}) + 2 \text{H}^+(\text{aq}) + 2 \text{e}^-$	$\longrightarrow \text{H}_2\text{O}_2(\text{aq})$	0.70
$\text{I}_2(\text{s}) + 2 \text{e}^-$	$\longrightarrow 2 \text{I}^-(\text{aq})$	0.54
$\text{O}_2(\text{g}) + 2 \text{H}_2\text{O}(\text{l}) + 4 \text{e}^-$	$\longrightarrow 4 \text{OH}^-(\text{aq})$	0.40
$\text{Cu}^{2+}(\text{aq}) + 2 \text{e}^-$	$\longrightarrow \text{Cu}(\text{s})$	0.34
$\text{Sn}^{4+}(\text{aq}) + 2 \text{e}^-$	$\longrightarrow \text{Sn}^{2+}(\text{aq})$	0.15
$2 \text{H}^+(\text{aq}) + 2 \text{e}^-$	$\longrightarrow \text{H}_2(\text{g})$	0
$\text{Pb}^{2+}(\text{aq}) + 2 \text{e}^-$	$\longrightarrow \text{Pb}(\text{s})$	-0.13
$\text{Ni}^{2+}(\text{aq}) + 2 \text{e}^-$	$\longrightarrow \text{Ni}(\text{s})$	-0.26
$\text{Cd}^{2+}(\text{aq}) + 2 \text{e}^-$	$\longrightarrow \text{Cd}(\text{s})$	-0.40
$\text{Fe}^{2+}(\text{aq}) + 2 \text{e}^-$	$\longrightarrow \text{Fe}(\text{s})$	-0.45
$\text{Zn}^{2+}(\text{aq}) + 2 \text{e}^-$	$\longrightarrow \text{Zn}(\text{s})$	-0.76
$2 \text{H}_2\text{O}(\text{l}) + 2 \text{e}^-$	$\longrightarrow \text{H}_2(\text{g}) + 2 \text{OH}^-(\text{aq})$	-0.83
$\text{Al}^{3+}(\text{aq}) + 3 \text{e}^-$	$\longrightarrow \text{Al}(\text{s})$	-1.66
$\text{Mg}^{2+}(\text{aq}) + 2 \text{e}^-$	$\longrightarrow \text{Mg}(\text{s})$	-2.37
$\text{Na}^+(\text{aq}) + \text{e}^-$	$\longrightarrow \text{Na}(\text{s})$	-2.71
$\text{Li}^+(\text{aq}) + \text{e}^-$	$\longrightarrow \text{Li}(\text{s})$	-3.04

Weaker
reducing
agent



Stronger
reducing
agent

Why 'Lithium'

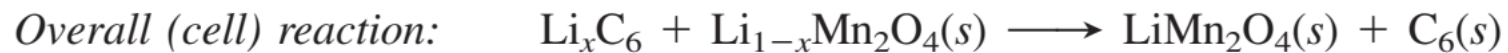
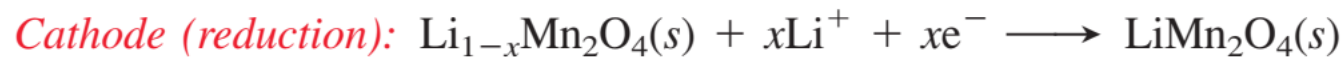


LITHIUM

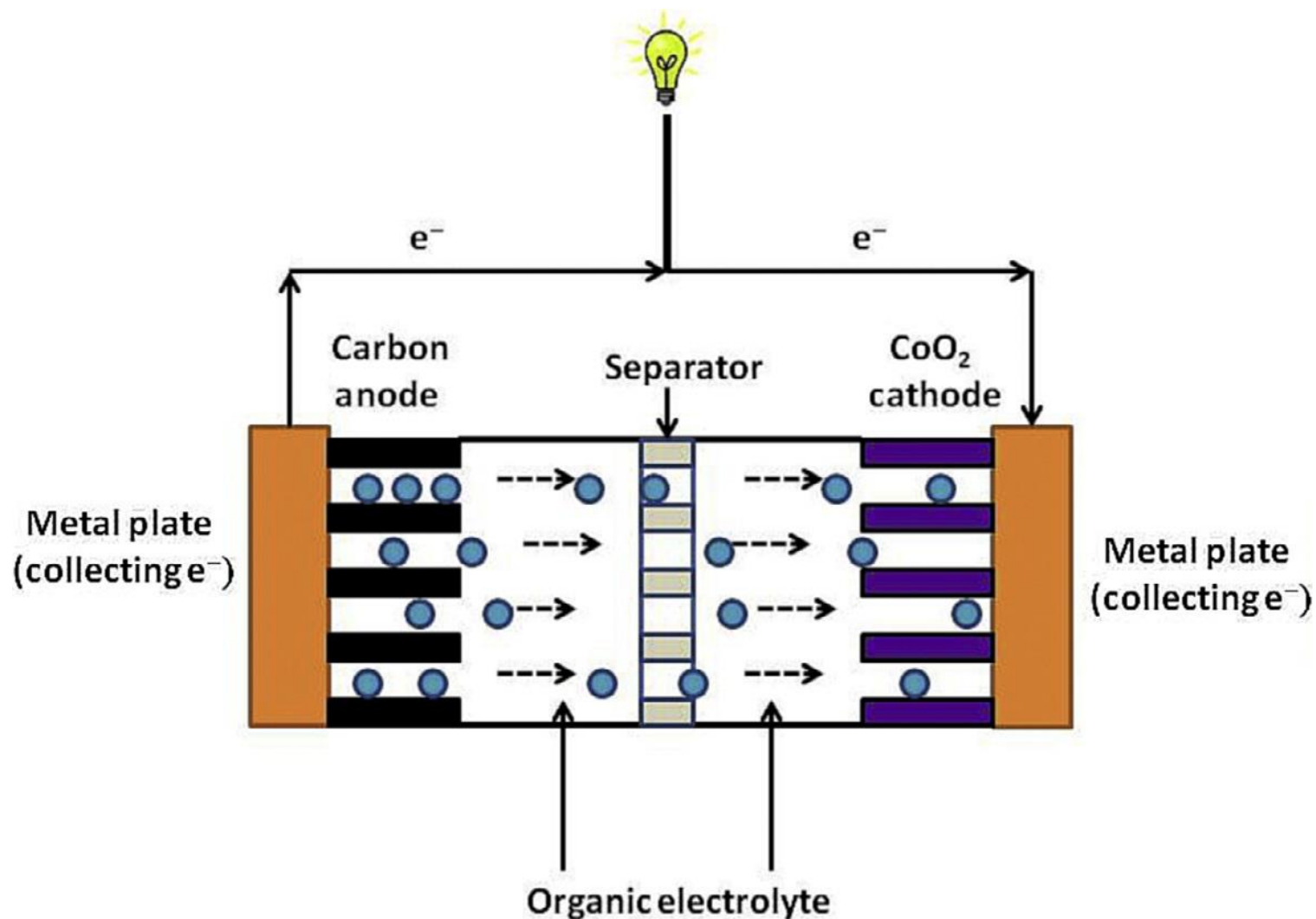
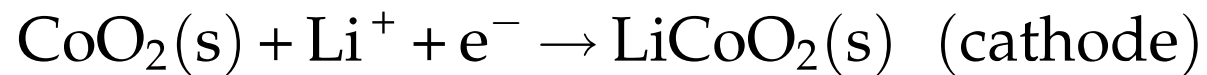
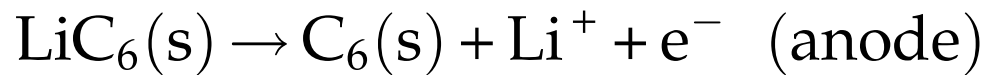
You Tube / LearnEngineering

<https://www.youtube.com/watch?v=VxMM4g2Sk8U>

- Cell reactions (discharge):



$$E_{\text{cell}} = 3.7 \text{ V}$$



Advantages of Li-ion battery

- ***High energy density:*** The high energy density is one of the chief advantages of lithium ion battery technology. With electronic equipment such as mobile phones needing to operate longer between charges while still consuming more power, there is always a need to batteries with a much higher energy density.
- ***Self-discharge:*** One issue with many rechargeable batteries is the self-discharge rate. Lithium ion cells is that their rate of self-discharge is much **lower than that of other rechargeable cells** such as Ni-Cad forms.
- ***Low maintenance:*** One major lithium ion battery advantage is that they do not require and maintenance to ensure their performance.
- ***No requirement for priming:*** Some rechargeable cells need to be primed when they receive their first charge. There is no requirement for this with lithium ion cells and batteries.
- ***Variety of types available:*** There are several types of lithium ion cell available. This advantage of lithium ion batteries can mean that the right technology can be used for the particular application needed.

Advantages of Li-ion Battery

- High energy density - potential for yet higher capacities.
- Does not need prolonged priming when new. One regular charge is all that's needed.
- Relatively low self-discharge - self-discharge is less than half that of nickel-based batteries.
- Low Maintenance - no periodic discharge is needed; there is no memory.
- Specialty cells can provide very high current to applications such as power tools

Disadvantages of Li-ion battery

- ***Protection required:*** Lithium ion cells and batteries are not as robust as some other rechargeable technologies. They require protection from being over charged and discharged too far.
- ***Ageing :*** One of the major lithium ion battery disadvantages for consumer electronics is that lithium ion batteries suffer from ageing.
- ***Transportation:*** This Li-ion battery disadvantage has come to the fore in recent years. Many airlines limit the number of lithium ion batteries they take, and this means their transportation is limited to ships.
- ***Cost:*** A major lithium ion battery disadvantage is their cost. Typically they are around 40% more costly to manufacture than Nickel cadmium cells.
- ***Immature technology:*** Although lithium ion batteries have been available for many years, it can still be considered an immature technology by some as it is very much a developing area.

Limitations of Li-ion Battery

- Requires protection circuit to maintain voltage and current within safe limits.
- Subject to aging, even if not in use - storage in a cool place at 40% charge reduces the aging effect.
- Transportation restrictions - shipment of larger quantities may be subject to regulatory control. This restriction does not apply to personal carry-on batteries.
- Expensive to manufacture - about 40 percent higher in cost than nickel-cadmium.
- Not fully mature - metals and chemicals are changing on a continuing basis.

Key Challenges in Battery Technology

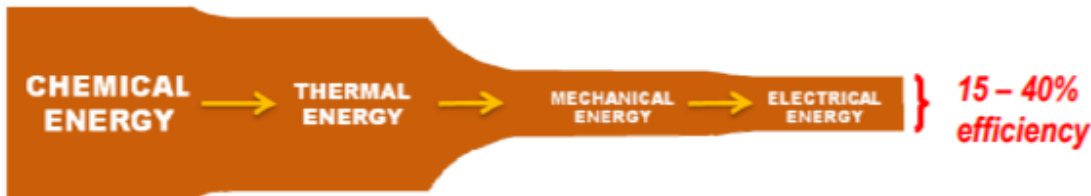
- *Means to increase energy density*
- *Means of increasing safety*
- *Means of reducing costs*
- *Means to achieve sustainable and greener Li-ion batteries*
- *Means of increasing capacity while ensuring sustainability and green storage*

Fuel Cells

Fuel Cells

Fuel cells convert chemical energy directly to electrical energy — with very high efficiency — and without criteria pollutant emissions.

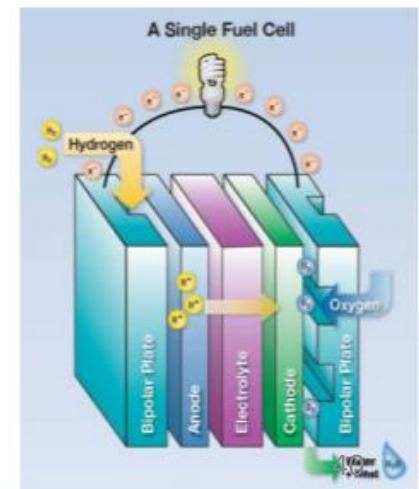
Energy Conversion in Combustion Engines



Energy Conversion Fuel Cells

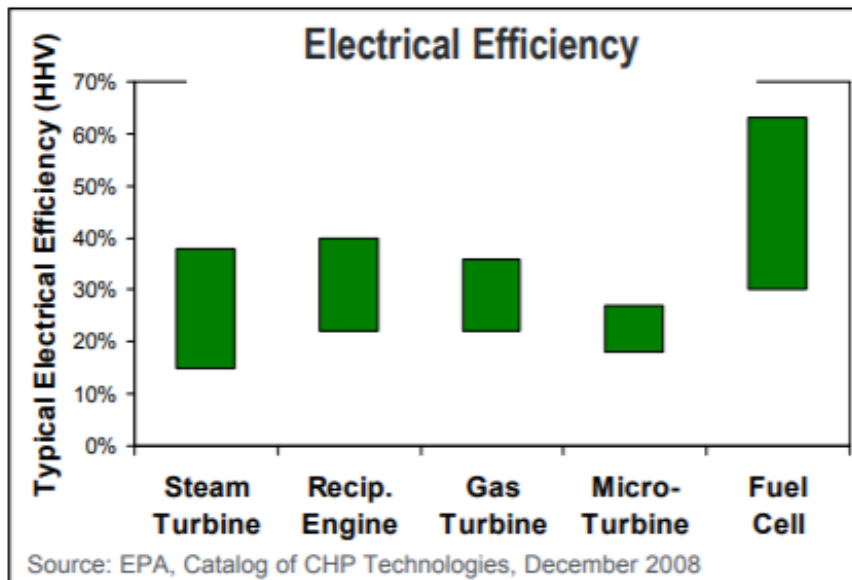


Fuel cells convert chemical energy directly into electrical energy, bypassing inefficiencies associated with thermal energy conversion

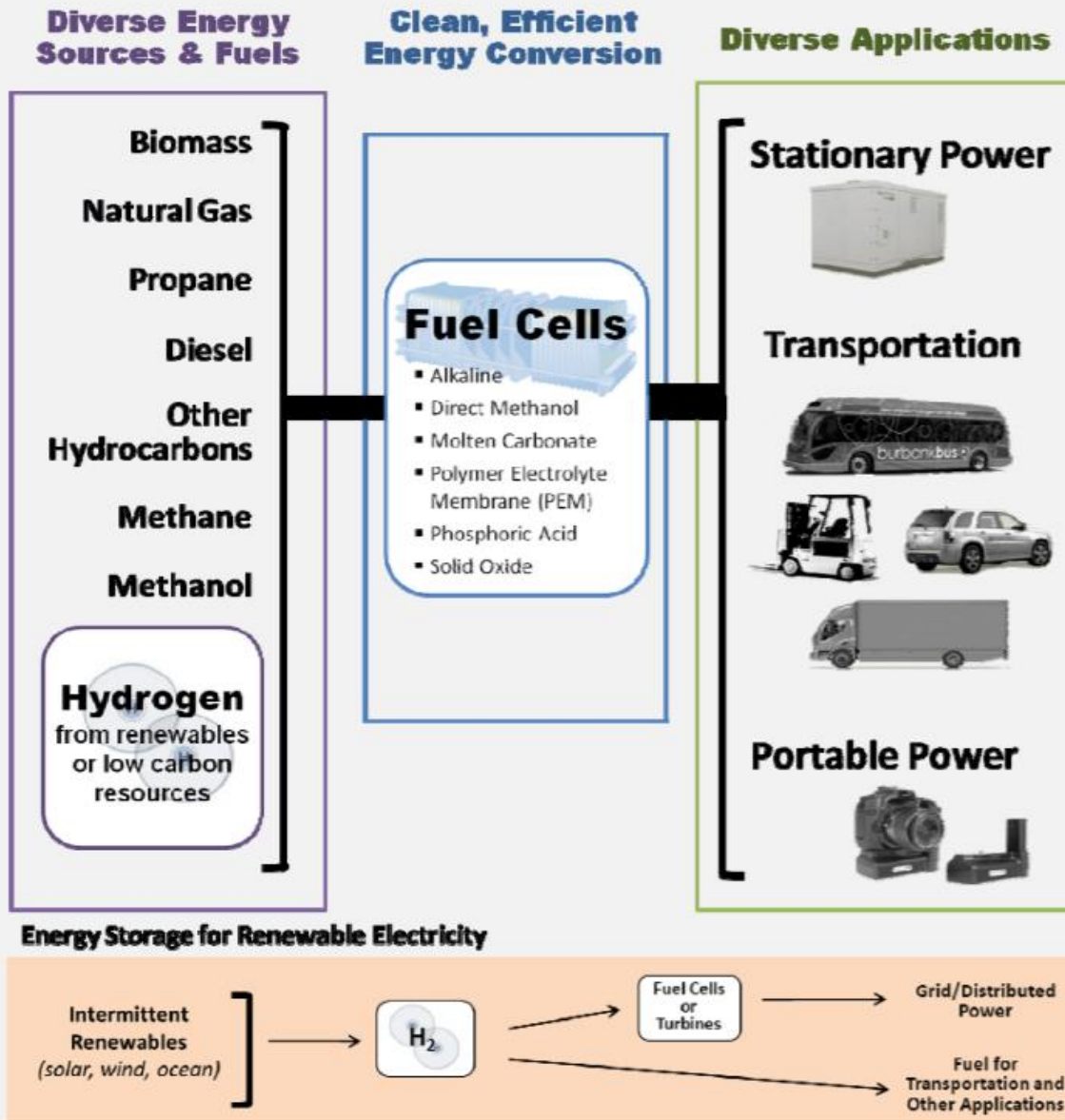


Combustion Engines — convert chemical energy into thermal energy and mechanical energy, and then into electrical energy.

Fuel cells — convert chemical energy directly into electrical energy, bypassing inefficiencies associated with thermal energy conversion. Available energy is equal to the Gibbs free energy.



The Role of Fuel Cells



Key Benefits

Very High Efficiency

- **> 60%** (*electrical*)
- **> 70%** (*electrical, hybrid fuel cell / turbine*)
- **> 80%** (*with CHP*)

Reduced CO₂ Emissions

- **35–50%+ reductions** for CHP systems (*>80% with biogas*)
- **55–90% reductions** for light-duty vehicles

Reduced Oil Use

- **>95% reduction** for FCEVs (*vs. today's gasoline ICEVs*)
- **>80% reduction** for FCEVs (*vs. advanced PHEVs*)

Reduced Air Pollution

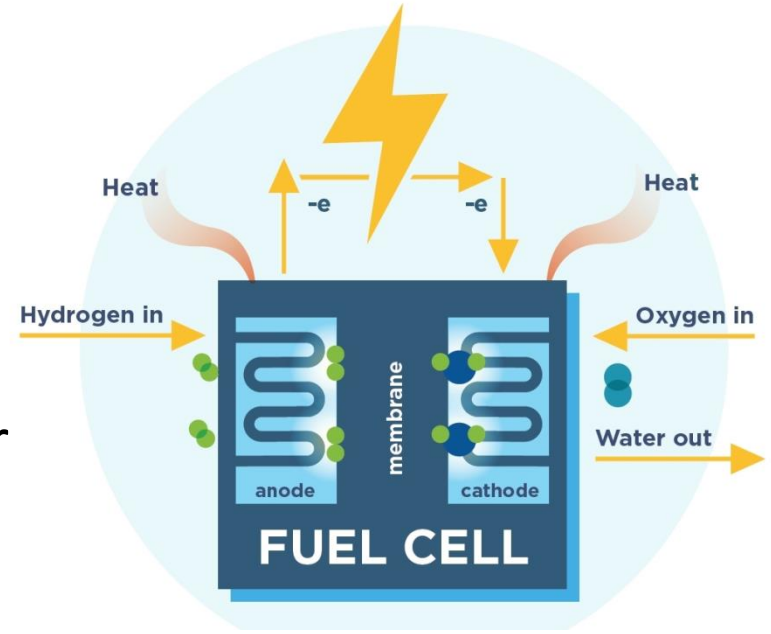
- **up to 90% reduction in criteria pollutants** for CHP systems

Fuel Flexibility

- **Clean fuels** — *including biogas, methanol, H₂*
- **Hydrogen** — *can be produced cleanly using sunlight or biomass directly, or through electrolysis, using renewable electricity*
- **Conventional fuels** — *including natural gas, propane, diesel*

Fuel Cells

- In contrast to primary and secondary batteries, a **fuel cell**, sometimes called a *flow battery*, is not self-contained.
- The reactants (usually a **combustible fuel** and oxygen) enter the cell, and the products leave the cell, generating electricity through the **controlled oxidation** of the fuel.
- In other words, *fuel cells use combustion to produce electricity*.
- The most common fuel cell being developed for use in cars is the *proton exchange membrane (PEM) cell*, which uses H_2 as the fuel and has an operating temperature of around $80\text{ }^{\circ}\text{C}$.

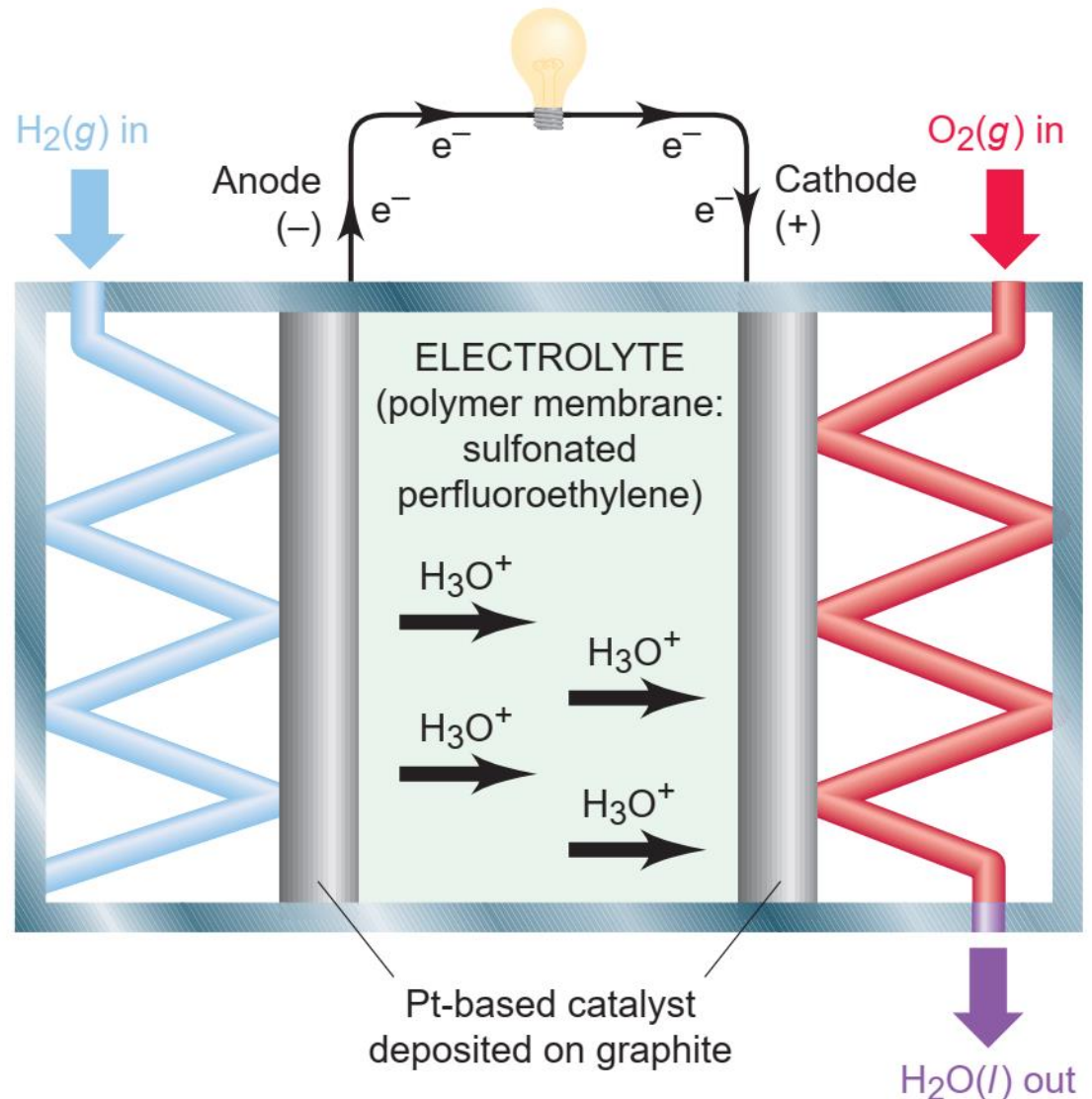


Types of Fuel Cells

We will be discussing the following fuel cells:

- Proton Exchange Membrane Fuel Cells (**PEMFC**)
- Solid Oxide Fuel Cells (**SOFC**)

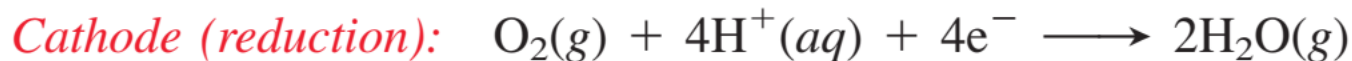
Proton Exchange Membrane Fuel Cells (PEMFC) are also called **Polymer membrane Fuel Cells**



Proton Exchange Membrane Fuel Cells (PEMFC)

- The cell reactions are.

Anode (oxidation):



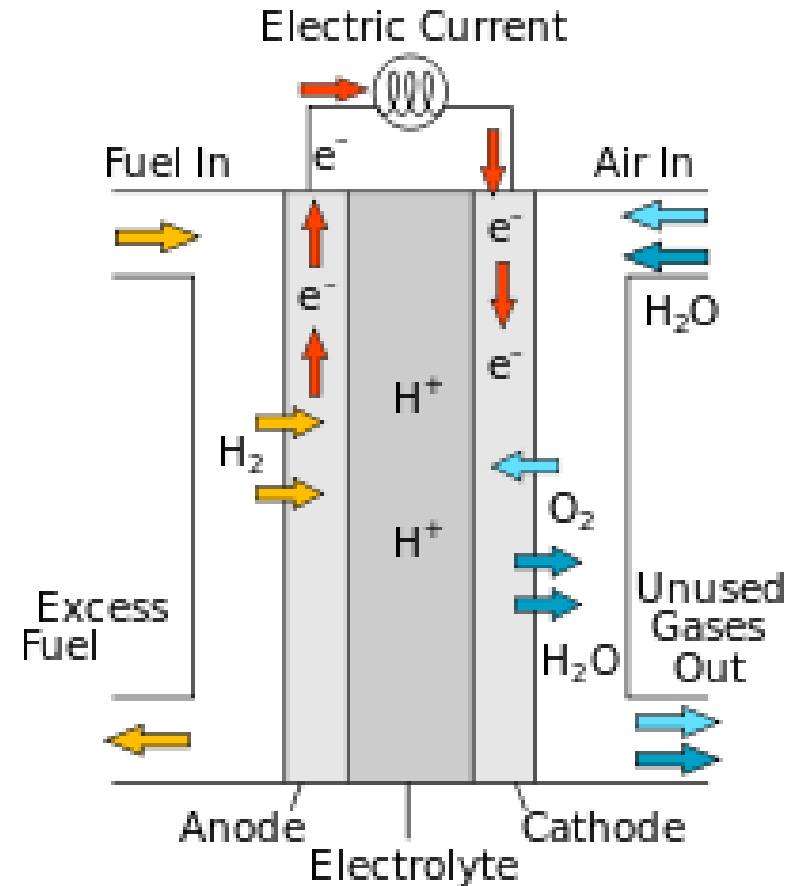
- The reactions in fuel cells have much lower rates than those in other batteries, so they require an *electrocatalyst* to decrease the activation energy.
- These are embedded in a polymer electrolyte membrane having a perfluoroethylene backbone ($-\text{[F}_2\text{C}-\text{CF}_2]-$) with n attached sulfonic acid groups (RSO_3^-) that play a key role in ferrying protons from anode to cathode.

Anode: Graphite or Pt/C

H_2 through anode : Oxidation

Cathode:

O_2 through cathode: Reduction

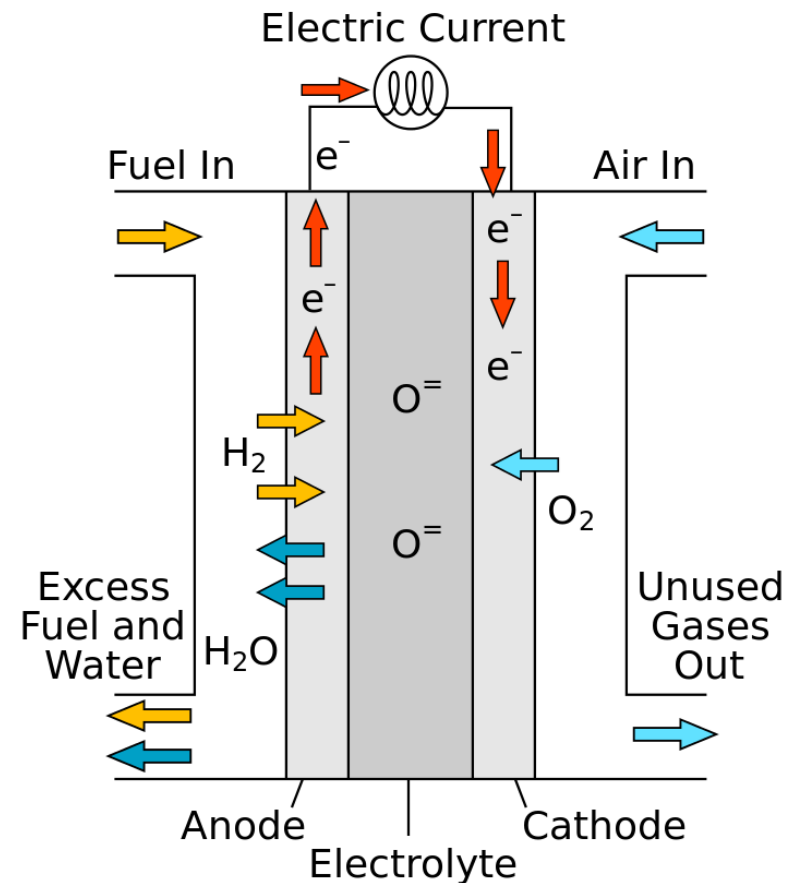


Applications of PEMFC

- Hydrogen fuel cells have been used for years to provide electricity and pure water during space flights.
- In the very near future, similar ones will supply electric power for transportation, residential, and commercial needs. Already, every major car manufacturer has a fuel-cell prototype.
- By themselves, these cells produce no pollutants, and they convert about 75% of the fuel's bond energy into useable power, in contrast to 40% for a coal-fired power plant and 25% for a gasoline-powered car engine.

Solid oxide fuel cell

- A **solid oxide fuel cell** (or **SOFC**) is an electrochemical conversion device that produces electricity directly from oxidizing a fuel.
- Fuel cells are characterized by their electrolyte material; the SOFC has a **solid oxide** or **ceramic electrolyte**



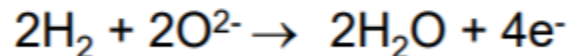
SOFC

- The electrolyte consists of a solid, nonporous metal oxide, typically Y_2O_3 stabilized ZrO_2 with the anode made from CoZrO_2 or NiZrO_2 cermet, while the cathode is made from Sr doped LaMnO_3 .
- The cell operates at 1200 to 1830 deg F or 650 to 1000 deg C such that conduction by oxygen ions through the electrolyte may occur.

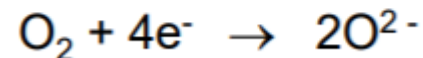
SOFC

- The electrochemical reactions in a typical solid oxide fuel cell fuelled with Hydrogen:

Reaction at the anode;



Reaction at the cathode;

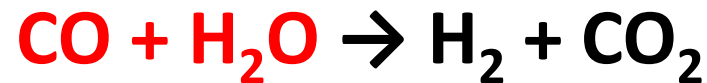


Overall electrochemical reaction for SOFC:

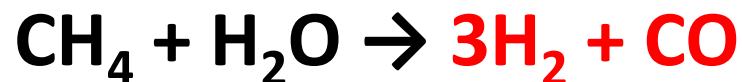


SOFC

- CO and hydrocarbons such as CH₄ can also be used as fuels in an SOFC.
- At the high temperatures within the cell, it is feasible for the **water gas** shift reaction:



and the steam reforming reaction:



(in the case of natural gas)
to take place to produce H₂ that is easily oxidized at the anode.

Advantages of Fuel Cell

- **High Efficiency**- when utilizing co-generation, fuel cells can attain over 80% energy efficiency
- **Good reliability**- quality of power provided does not degrade over time.
- **Noise**- offers a much more silent and smooth alternative to conventional energy production.
- **Environmentally beneficial**- greatly reduces CO₂ and harmful pollutant emissions.
- **Size reduction**- fuel cells are significantly lighter and more compact

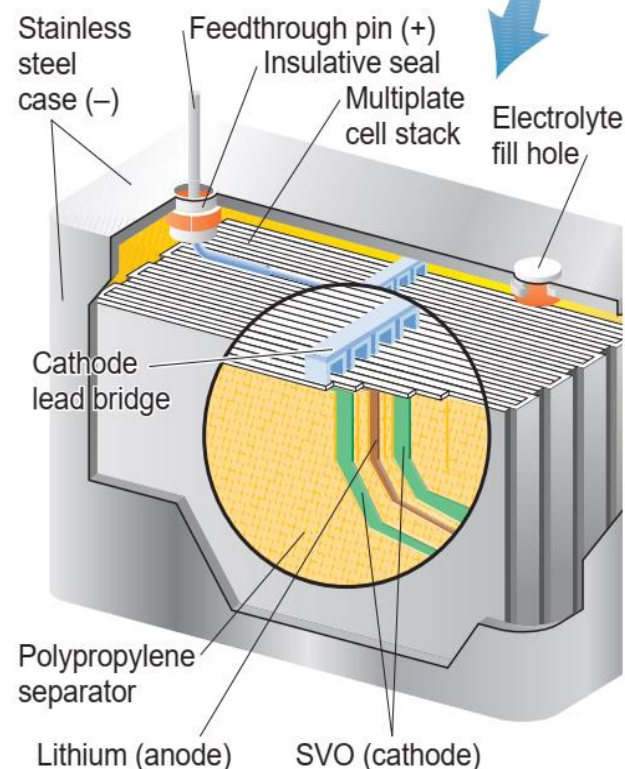
Disadvantages of Fuel Cell

- **Expensive to manufacture** due the high cost of catalysts (**platinum**)
- Lack of infrastructure to support the distribution of hydrogen
- A lot of the currently available fuel cell technology is in the prototype stage and not yet validated.
- **Hydrogen** is expensive to produce and not widely available

Special Notes:

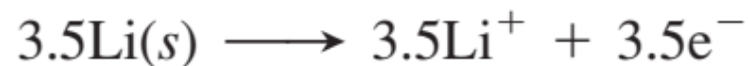
Primary Lithium Battery

- The primary lithium battery is widely used in watches and implanted medical devices. It offers an extremely **high energy/mass ratio**, producing 1 mol of e^- (1 F) from less than 7 g of metal.
- The anode is lithium metal foil, which requires a nonaqueous electrolyte.
- The cathode is one of several metal oxides in which lithium ions lie between oxide layers. Some implantable medical devices have a silver vanadium oxide (SVO; $\text{AgV}_2\text{O}_{5.5}$) cathode and can provide power for several years, but at a low rate because energy storage is limited



- The cell reactions of lithium battery:

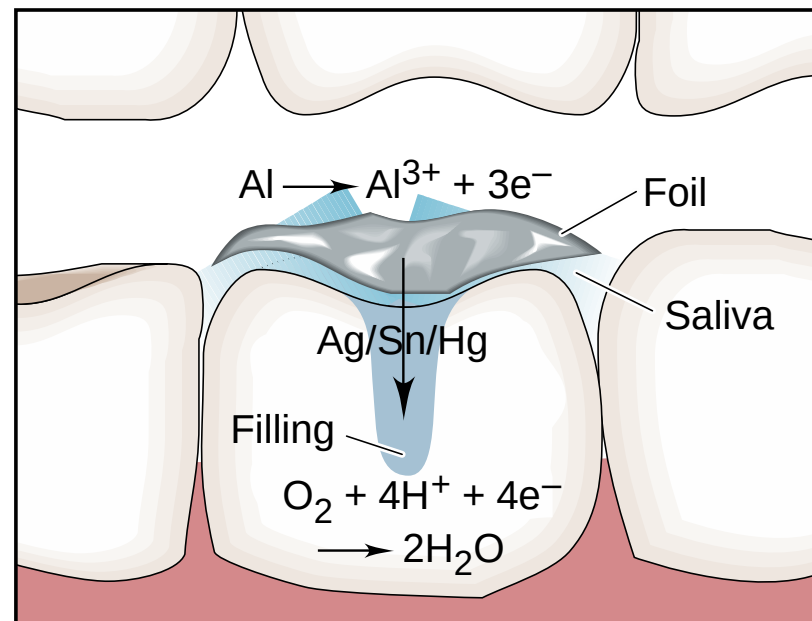
Anode (oxidation):



Cathode (reduction): $\text{AgV}_2\text{O}_{5.5} + 3.5\text{Li}^+ + 3.5\text{e}^- \longrightarrow \text{Li}_{3.5}\text{AgV}_2\text{O}_{5.5}$

Overall (cell) reaction: $\text{AgV}_2\text{O}_{5.5} + 3.5\text{Li}(s) \longrightarrow \text{Li}_{3.5}\text{AgV}_2\text{O}_{5.5}$

Special Notes:

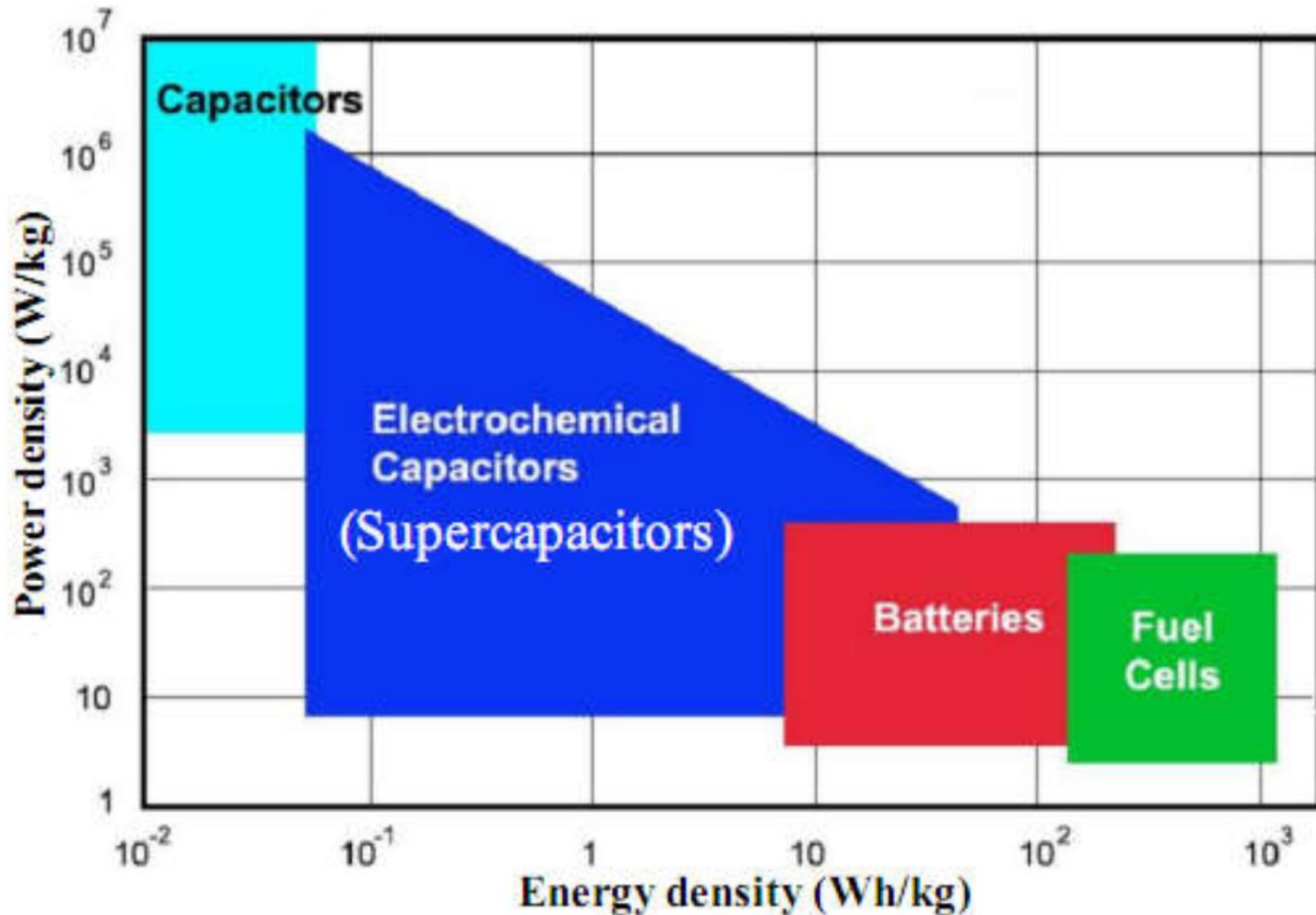


● The Pain of a Dental Voltaic Cell

Have you ever felt a jolt of pain when biting down with a filled tooth on a scrap of foil left on a piece of food? Here's the reason. The aluminum foil acts as an active anode (E° of Al = -1.66 V), saliva as the electrolyte, and the filling (usually a silver/tin/mercury alloy) as an inactive cathode. O_2 is reduced to water, and the short circuit between the foil in contact with the filling creates a current that is sensed by the nerve of the tooth.

Ragone plot

- Energy density vs Power density plot



Capacitors



Capacitor

- A **capacitor** is a device that **stores electrical energy** in an electric field.
- It consist of Cathode and Anode, no electrolyte needed
- The capacitance is measured in units of Farad (F).

Capacitors Explained

How does a capacitor work?

TheEngineeringMindset.com

<https://www.youtube.com/watch?v=X4EUwTwZ110>



<https://www.youtube.com/watch?v=5hFC9ugTGLs>

Supercapacitor

- A **capacitor** is a device that stores electrical energy in an electric field.
- A **supercapacitor (SC)**, also called an **ultracapacitor**, is a **high-capacity capacitor** with a capacitance value much higher than other capacitors, but with **lower voltage limits**, that bridges the gap between electrolytic capacitors and rechargeable batteries.
- Supercapacitors have similarities with both batteries and conventional capacitors.
- Unlike batteries, SC's store electrical energy, not chemical energy. Unlike capacitors SC's contain moving ions.
- They are also called - ***electrochemical capacitor*** (EC Capacitor), Electric double-layer capacitor (EDLC), ultracapacitor,

Rechargeable Batteries Vs Supercapacitor

Batteries

- ❖ Have higher energy density
- ❖ Typically 200–1000 charge-discharge cycles
- ❖ Contain highly reactive and hazardous chemicals
- ❖ Negatively effected by low temperatures

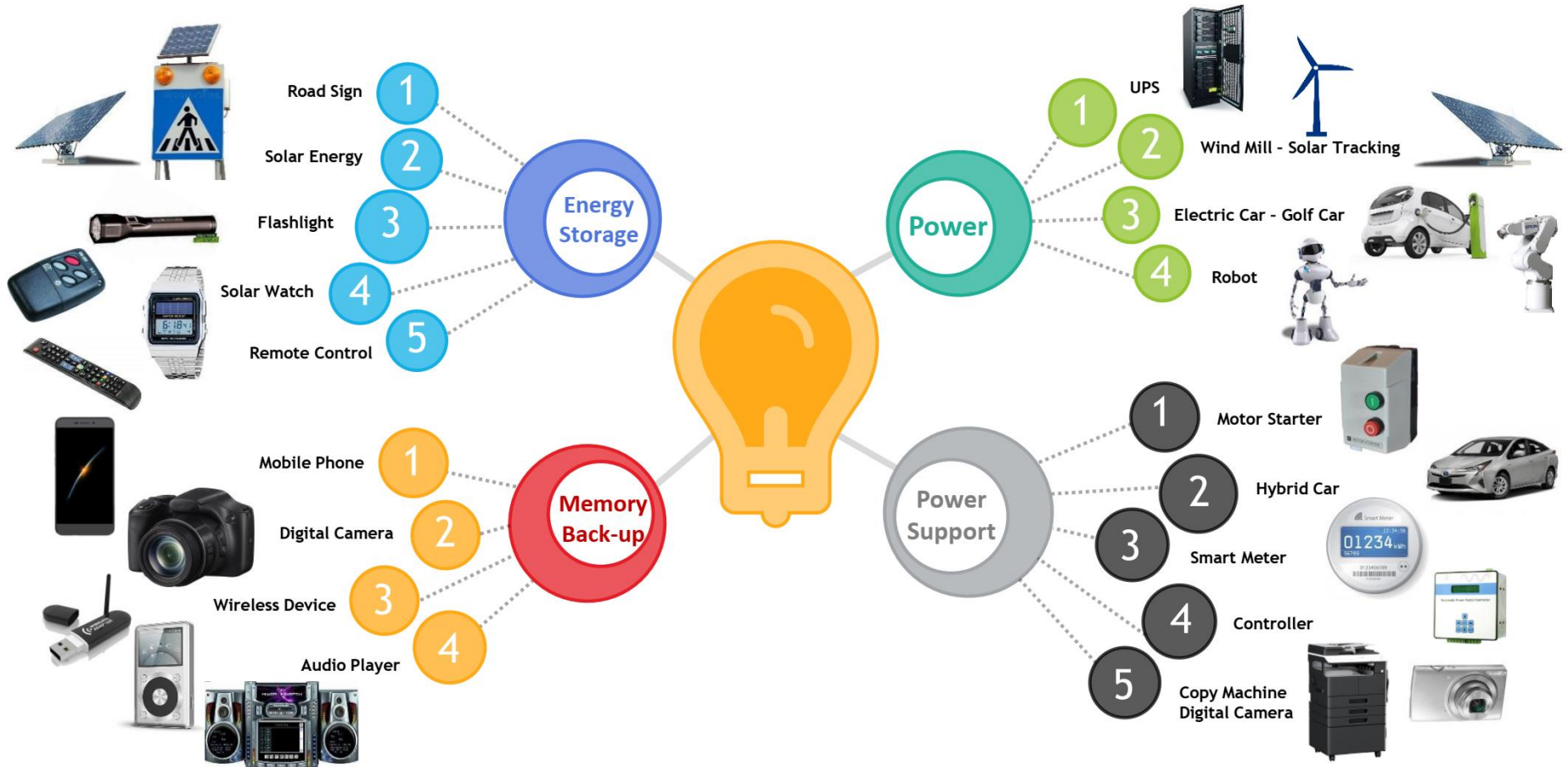


Supercapacitors

- ❖ Higher power density
- ❖ Much faster charge and discharge rate
- ❖ Environmentally friendly
- ❖ Extremely low internal resistance or ESR
- ❖ High efficiency (97-98%)
- ❖ Over a million charge-discharge cycles

Applications of Supercapacitor

- Automobiles, Trains, cranes, elevators, etc
- Consumer electronics (Laptop, digital cameras, etc)
- Tools (cordless electric screwdriver)
- Grid power buffer
- Voltage stabilizer
- Street light
- Transport



Specialties of supercapacitor

- Supercapacitors (SC) can be fully charged and discharged **in seconds**
- SC's can be charged and discharged even up to a million times.
- SC's have very **low energy densities** vs. batteries;
- SC's have very **high self-discharge** vs. batteries.
- Large amount of power in relatively short time
- **Long shelf life** as compared to batteries. In batteries, the energy stored and released as in the form of chemical reaction, which causes degradation
- SC can **recharge in short time**, and supply high and frequent power demand peaks
- SC have **high power density** and can provide large power burst for short duration

Next generation Battery!!!

Indian Scientists Develop High Performance Hybrid Supercapacitors For Next-Gen Energy Storage Devices

By Swarajya Staff

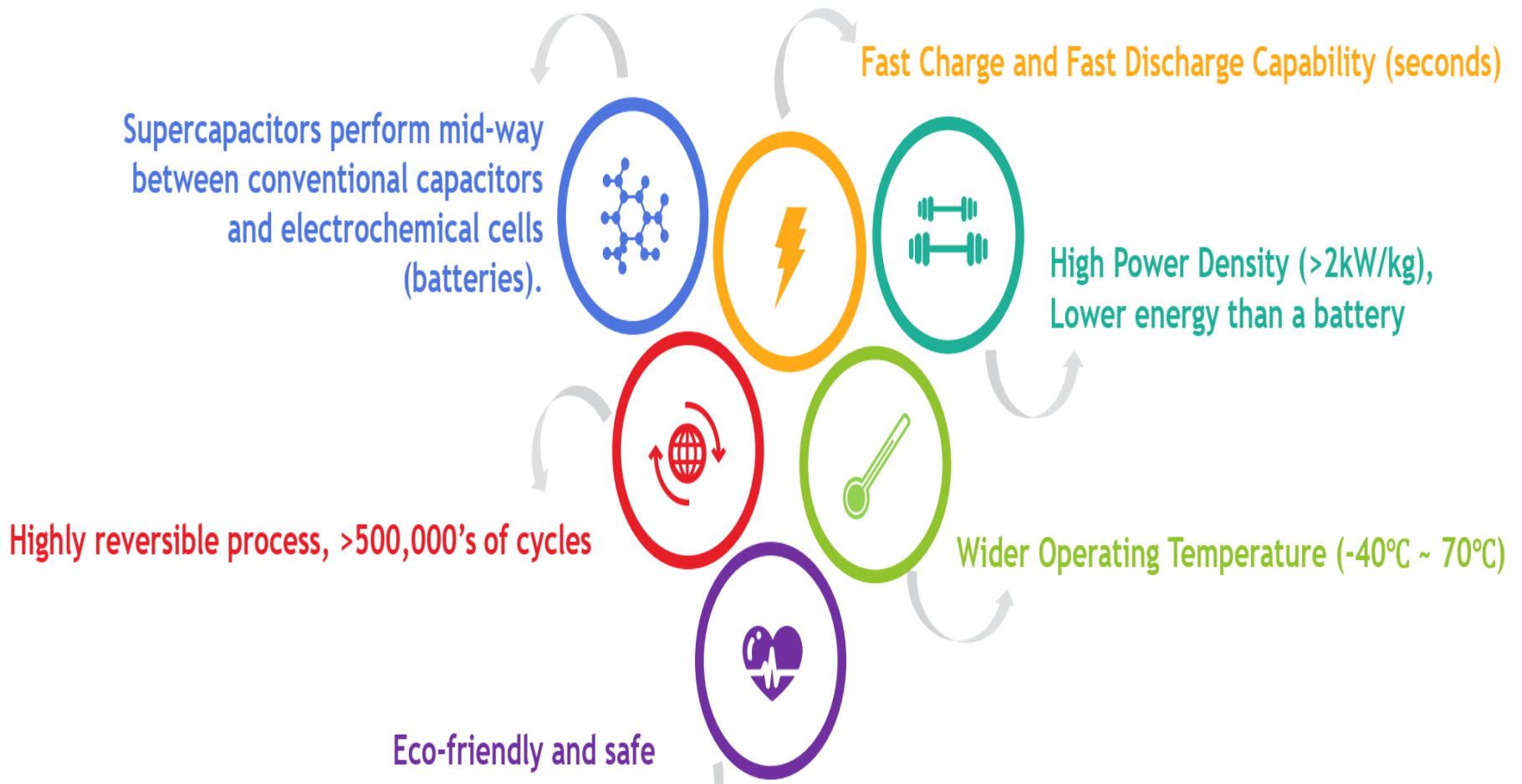
December 17, 2020 at 4:37 PM



Indian scientists have developed a low-cost supercapacitor device with excellent capacitive retention using a novel electrode material they have synthesized. The research can pave the way for the next generation high power-high energy storage devices.

Supercapacitors have gained considerable attention due to their high power density, long cycle life, and excellent capacity retention compared to their battery counterparts.

Supercapacitors with high capacitance and excellent capacitive retention developed from low-cost fabrication techniques are the need of the hour, considering their potential utility in the



Ragone plot

- Energy density vs Power density plot

